

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

November – December 2011

EC- 203 Solid State Electronics

Date: 01.12.2011, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Draw transistor amplifier in its h-parameter model. Derive equation of current gain and voltage gain 245 [4]

(b) Illustrating neat diagram and describe the operation of photo diode. [3]

Q-2 (a) Design a Voltage divider bias circuit for the specified conditions. $V_{CC} = 12V$, $V_{CE} = 6V$, $I_C = 1mA$, $S = 20$, $\beta = 100$ and $V_E = 1V$. [7]

(b) Draw the Hybrid- Π model at low frequencies. Derive input conductance, feedback conductance and base spreading resistance. 350 [7]

OR

Q-2 (a) In the circuit shown in **Figure 1**, $R_B = 300 K\Omega$ and $R_E = 150 K\Omega$. Calculate I_B , I_C , V_{CE} and also determine the region of operation. [7]

(b) Explain Miller's theorem and it's Dual using neat sketches. 255 [7]

Q - 3 (a) Illustrate diagram of minority carrier charge distribution in base region. Explain diffusion capacitance for the transistor at high frequencies. [7]

(b) Consider a single stage CE amplifier with $R_s = 1 K\Omega$, $R_1 = 50K\Omega$, $R_2 = 2K\Omega$, $R_C = 1K\Omega$, $R_L = 1.2 K\Omega$, $h_{fe} = 50$, $h_{ie} = 1.1K$, $h_{oe} = 25 \mu A/V$ and $h_{re} = 0.00025$ as shown in **Figure 2**. Calculate Current gain, Input resistance and Voltage gain. [7]

OR

Q - 3 (a) What is thermal runaway in regard to BJT? Explain any one compensation technique. [6]

(b) Explain ripple factor. Draw & Explain Π -filter. Derive ripple factor of Π -filter. [4]

(c) Derive the stability factor (S) for Voltage divider bias. [4]

SECTION – II

- Q-4 (a)** The amplifier shown in **Figure 3** utilizes an n-channel FET for which $V_p = -2.0$ V and $I_{DSS} = 1.65$ mA. It is desired to bias the circuit at $I_D = 0.8$ mA using $V_{DD} = 24$ V. assume $r_d \gg R_d$ find (a) V_{GS} (b) g_m (c) R_s (d) R_d , such that the voltage gain is at least 20 dB with R_s bypassed with a very large capacitance C_s . [8]

- (b) Define the following types of distortion: Nonlinear, Frequency, Phase shift distortion [6]

OR

- Q-4 (a)** Find an expression for signal voltage across R_L . The two FETs are identical with parameters μ , r_d , g_m for the **Figure 4**. [7]

- (b) Explain Volt- ampere Characteristics of JFET. [4]

- (c) Describe RC coupled amplifier operation with necessary diagram [3]

- Q-5 (a)** Explain Mobility and Conductivity with necessary derivation. [4]

- (b) Why a semiconductor acts as an insulator at 0° K and why its conductivity increases with increasing temperature? [3]

- Q-6 (a)** Calculate the peak current in $12.5\text{K}\Omega$ resistor in circuit in **Figure 5**. [4]

- (b) Define Rise time and Tilt/Sag with proper figure. [3]

- (c) Define the pinch off voltage V_p . Sketch and explain the depletion region before and after pinch-off. [7]

OR

- (a) Explain the Hall effect. [5]

- (b) Define Mean life time & mean free path [2]

- (c) Differentiate FET and BJT. Draw & Explain basic structure of n-channel FET. [7]

